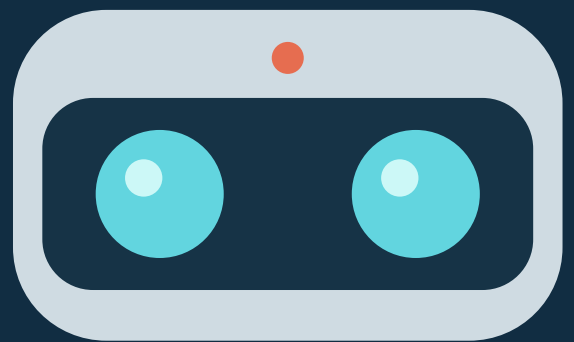


R.O.B. VISION

OPTICAL ROBOT COMPANION / MK.01



USER GUIDE

Play, preview, setup, and problem solving

CURRENT REFERENCE / 25 SEPTEMBER 2026

UNO Q + RETROPIE LIVE · TEST SIGNAL VERIFIED

CHAPTER 01 / CURRENT REFERENCE

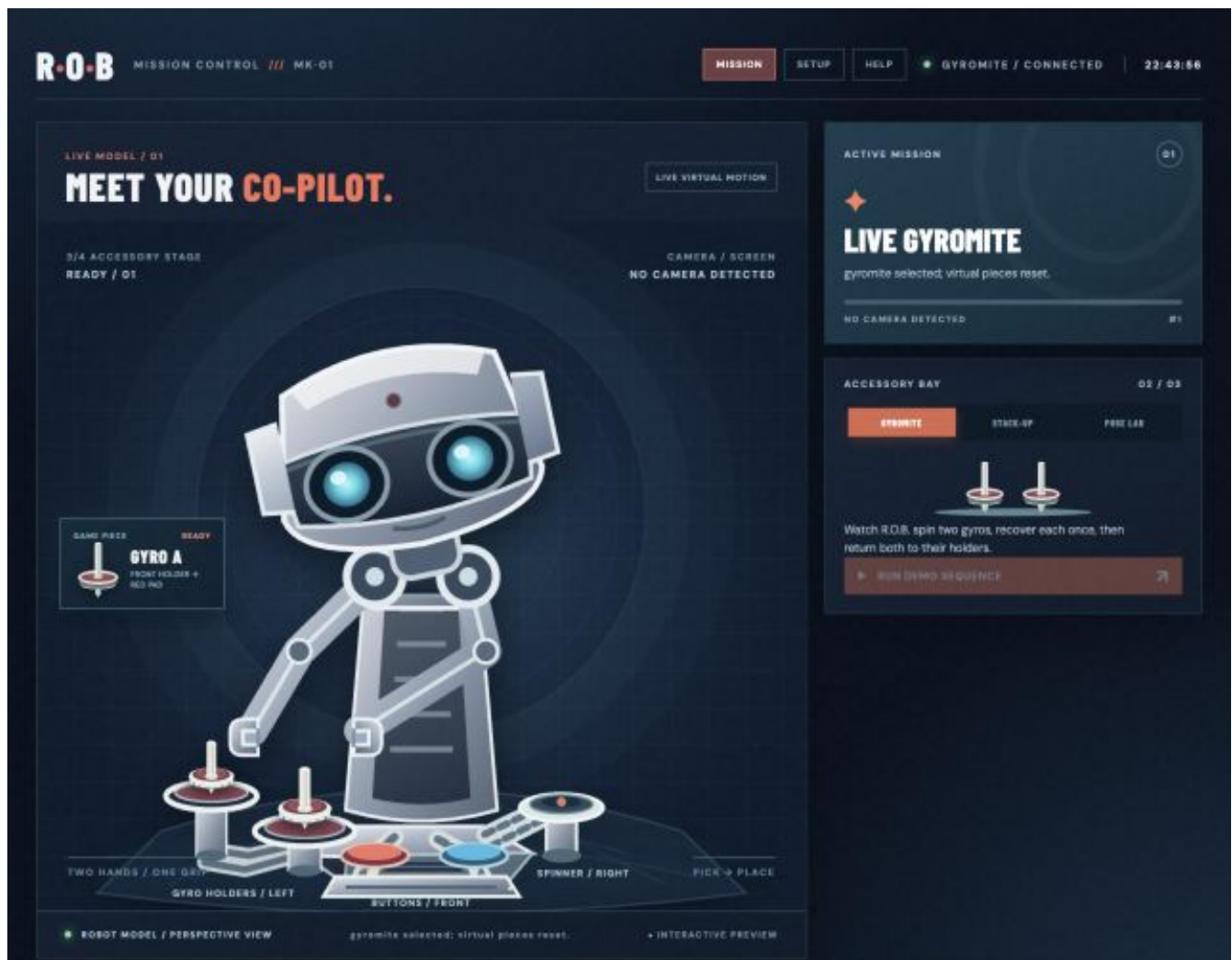
R.O.B. Vision user guide

Current release: The UNO Q hosts a live virtual R.O.B. dashboard and RetroPie link. Both Gyromite gate controls have been confirmed in Game A. The RetroPie game-frame link decoded all six commands in both supported games during live Direct play. The camera recognized both games' Test modes at about 60 fps; its movement reception was intermittent. R.O.B., gyros, blocks, trays, and spinner are graphics and software state, not physical objects.

Open R.O.B. Vision

Start **R.O.B. Vision** in UNO Q App Lab, then open `http://arduinoain.local/dashboard/` on a laptop, phone, or tablet. App Lab runs one app at a time on this UNO Q; stop VirtualGlove before starting R.O.B. Vision. The direct `http://arduinoain.local:8766/dashboard/` address reaches the same controller. A page opened from `file://` is an offline preview and does not show live RetroPie state.

Mission keeps the animated R.O.B. as the main view. Accessory Bay and Game Table show his virtual game pieces. Pose Preview and System Vitals sit beside the Game Table; the Activity Feed reports controller actions. A label such as **GYROMITE / CONNECTED** means Gyromite is selected and this browser can reach the UNO Q. RetroPie status appears separately as **RETROPIE ONLINE** or **RETROPIE OFFLINE**. **GAME FRAMES LINKED** means the selected game is actively sending frames through the RetroPie link.



Mission on the UNO Q, with Gyromite selected. The camera was not attached when this screenshot was taken.

Try the demonstration

When no supported game is running, select **Gyromite** or **Stack-Up** and **Run Demo Sequence**. It works even if RetroPie is paired but idle. Gyromite demonstrates a held unspun gyro, two spinning gyros on pads, spin-down, re-spin, and return to holders. The illustrative spin clock is 55 seconds. Stack-Up moves red alone and then a blue/white group. **Home** resets the pieces. Pose Lab is a character-only movement sandbox. A live game launch stops the demo and selects that game's scene.

Choose an NES emulator

For automatic R.O.B. movement on RetroPie, select **lr-robvision-fceumm** if you prefer FCEUmm, or **lr-robvision-nestopia** if you prefer Nestopia. These are launch choices that use the corresponding original emulator core while sending its game flashes to the UNO Q. Both choices are installed on the test RetroPie; Gyromite and Stack-Up currently default to the FCEUmm choice. The plain **lr-fceumm** and **lr-nestopia** choices play normally but do not send game frames to R.O.B. Vision. The camera remains an experimental way to read those flashes from the screen. See the [RetroPie setup guide](#) to change a game's saved choice.

Play Gyromite

Open [Setup](#) to check the RetroPie link and, when a camera is attached, frame the game display. Start Gyromite on RetroPie and watch Mission for **GAME FRAMES LINKED**. R.O.B. then follows complete light commands from the game's rendered frames. On Mission, **Fast Gates** works while Gyromite is selected and camera capture is stopped. Tap **Lower Blue** or **Lower Red** to press immediately, tap again to release, or select **Release Both**. Keyboard shortcuts are **2**, **1**, and **0**. Presses expire after 60 seconds. Both buttons release on game exit or lost receiver data. If you moved a gyro with normal controls, select **Home** before Fast Gates to restore their holders.

A spinning virtual gyro on a pad keeps its gate pressed while R.O.B. moves elsewhere. An unspun gyro can press one gate while R.O.B. holds it there. The UNO Q sends the resulting pad states to RetroPie's virtual Controller 2. The player confirmed red and blue gate responses with the current mapping. The game screen itself remains the authority for Hector's position and gate animation.

Play Stack-Up

Stack-Up starts with all five colored blocks on Tray 3 in the current live model. In Direct mode, jumping Hector onto a command tile sends **Left**, **Right**, **Up**, **Down**, **Open**, or **Close** through the game-frame link. Manual controls use the same virtual model. A lower grip carries the contacted block and all those above it. Impossible moves are rejected and leave the blocks in place. The model does not currently initialize the distinct historical Bingo starting layout or judge a game's score. Stack-Up has no Gyromite-style Controller 2 gate path.

Camera and reset

On Setup, **Start Camera Check** shows the camera image, sampling rectangle, signal, and measured frame rate. **Watch Test Signal** blinks the virtual red light when a game's Test-mode signal is recognized; it does not move R.O.B. **Reset & Reconnect** closes and retries the camera in software. This does not power-cycle the USB hub. The camera capture code requests 60 fps. RetroPie game-frame play works without it; camera-only movement remains experimental.

Home resets the selected virtual game. Live **Emergency Stop** stops camera capture and clears the game selection; in the offline preview it cancels the scripted animation until reset. A browser reload restores the UNO Q's current snapshot. See [Setup](#), [gameplay](#), and [troubleshooting](#).

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R.O.B. Vision gameplay guide

Status: scripted preview and connected UNO Q controller. R.O.B., the gyros, spinner, pads, trays, and blocks exist only in the virtual model. The UNO Q controller owns their state and updates the browser. Red and blue pad presses reached RetroPie's virtual joystick, and the player confirmed both gate colors respond in Game A. The RetroPie frame link decoded all six command types in both games during live Direct play; camera-only timing remains experimental.

Gyromite

The original Gyromite booklet describes **Test**, **Direct**, **Game A**, and **Game B**. Direct sends six R.O.B. actions: up, down, left, right, open, and close. In Game A, Start switches into the robot-transmission screen for a command and then back to the professor. In Game B, the professor walks while Controller 1 directly commands R.O.B. A gyro on the blue or red tray lowers the matching gate.

The manual explicitly says a gyro **does not need to spin to operate only one gate**. R.O.B. can hold an unspun gyro on the virtual pad, then lift it to release that pad. If both gates must be controlled, spinning one gyro lets it stay upright on one pad while R.O.B. uses his hands to put the other on the other pad. A gyro that is no longer needed belongs on its holder. The spinner is a means to free R.O.B.'s hands, not an obligatory step for every pad press.

Available preview: the finite Gyromite demo shows an unspun held press, a two-gyro spin relay, separate spin-down and one recovery for each gyro, then a tidy return to both holders. It stops at COMPLETE. The spin lifetime is illustrative. Select **Home** to reset. In the connected setup, UNO Q pad state reaches RetroPie's virtual Controller 2; the browser's local-only preview does not drive that receiver.

Live controller: the UNO Q maintains the locations and spin phases of Gyro A and B. It derives red and blue virtual button states from those modeled conditions and sends them over LAN to the paired RetroPie Controller 2 receiver. It releases both buttons on game exit, connection loss, stale packets, or mode change. The player confirmed that both red and blue gate controls work in Game A with the corrected Controller 2 mapping. The **Fast Gates** dashboard controls place or return the matching virtual gyro and send the pad state immediately, with a 60-second automatic release. This keeps gate operation responsive while the player moves Hector. The RetroPie frame link controls R.O.B.'s virtual actions from complete rendered-frame commands, with the camera path available when the link is absent; the launch hook supplies the game ID. The controller keeps game context and modeled piece state distinct from unknown on-screen professor or gate positions.



Pose Preview, Fast Gates, and the simulated Game Table on Mission.

Stack-Up

The [Stack-Up booklet](#) describes Direct, Memory, and Bingo. The virtual fixture has five numbered trays and five colored blocks. Direct and Memory start with all five on Tray 3, top to bottom **red, white, blue, yellow, green**. One-player Bingo historically uses any color order on Tray 3; two-player Bingo starts with three on Tray 3 and one each on Trays 2 and 4. The current live model initializes the Direct/Memory stack for all modes; Bingo-specific layouts are not implemented. Direct moves Professor Hector onto six command keys: **left, right, up, down, open, close**. Memory plays a programmed command series; Bingo sends a command when one row or column completes. The UNO Q code applies each validated optical command as one virtual action; all six command types were observed through the live RetroPie frame link. It must model the grippers' height, so R.O.B. can carry a block together with all blocks above it. Stack-Up's documented modes do not need Gyromite's virtual Controller 2 pad return path, even though two-player Bingo uses a second human controller.

Available preview: R.O.B. first moves red alone from Tray 3 to Tray 4, then grips blue and white together and carries both to Tray 2. Green and yellow remain on Tray 3. All five blocks appear in the Live Model, Accessory Bay, and Game Table as they move. The **Stack-Up Commands** buttons above System Vitals apply one virtual left, right, up, down, open, or close action. **Home** restores the starting stack. A blocked move leaves all blocks in place and explains why. The preview demo is scripted; the separate live controller accepts manual commands and complete ROM-pattern detections from RetroPie game frames or optional camera capture. In game play, the player compares the virtual arrangement with the target shown by the game. The flash stream does not tell the controller which block to move or whether the target was achieved.

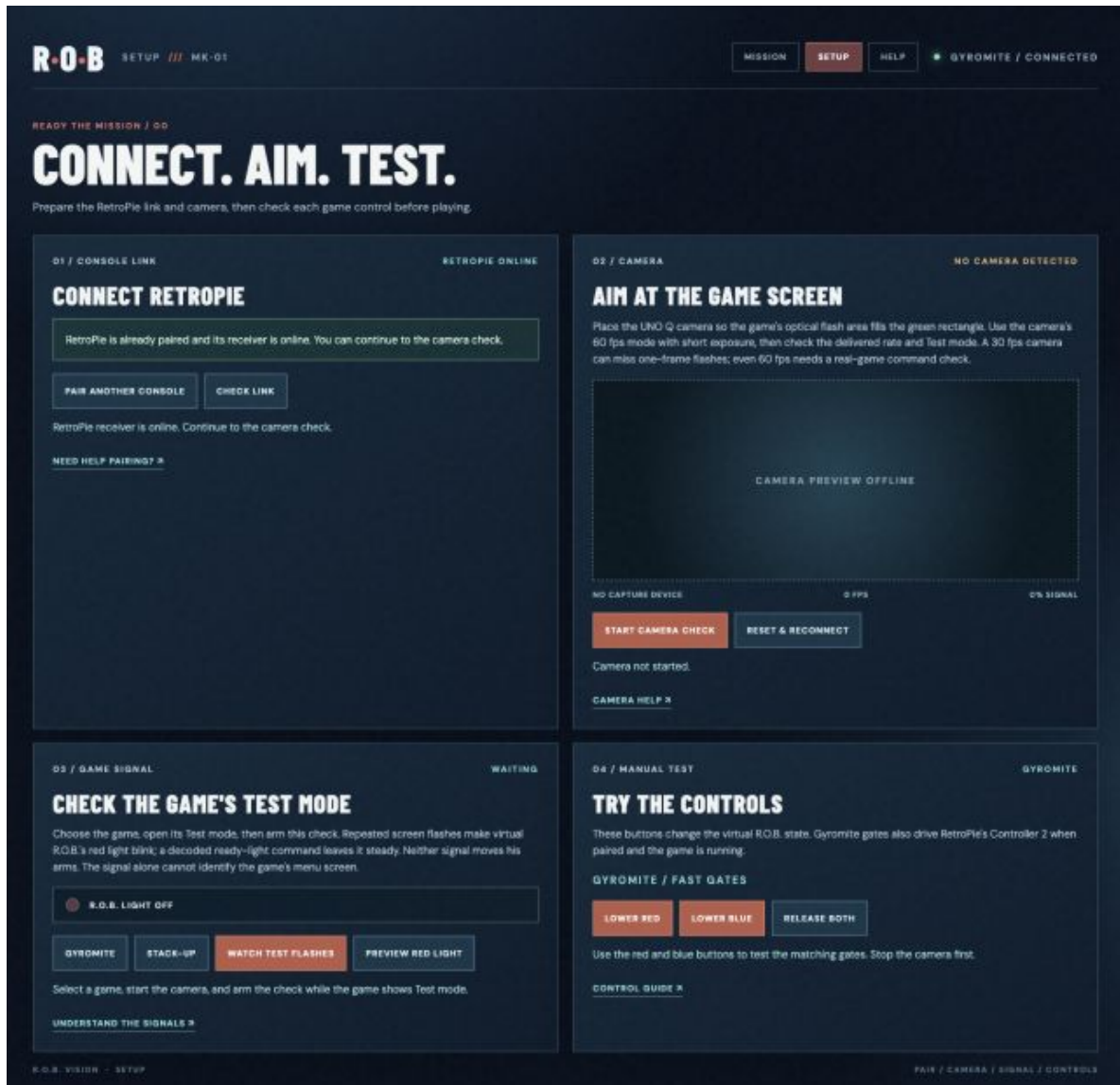
Pose Lab and recovery

Pose Lab exercises R.O.B.'s head, body turn, arms, and paired grip without game pieces or commands. Its demo is a short greeting; the controls below the scene let the viewer try poses manually. It is an art and motion sandbox, not a third game. In a live session, unreadable flashes produce no movement. A browser reconnect receives the UNO Q's current snapshot; it never restarts an action. If the Gyromite return path disconnects, virtual Controller 2 buttons release even if the browser still shows a gyro on a pad. The game host receives controller states, not a proof that a level was completed.

CHAPTER 03 / CURRENT REFERENCE

R.O.B. Vision Setup

Open `http://arduiain.local/dashboard/setup.html`. It opens directly to the RetroPie link status. If the receiver is already online, no pairing step is needed. Setup has four checks:



The Setup layout; this screenshot predates the connected camera test.

1. **Pair RetroPie.** The receiver and launch hooks must already be installed on RetroPie. Run `python3 /home/pi/rob-vision/tools/retropie_pair.py` in a RetroPie terminal. For five minutes it shows a six-digit code and a SHA-256 fingerprint. Enter its hostname, code, and fingerprint on Setup, then select **Pair Console**. The UNO Q checks the console certificate fingerprint before sending its token over the encrypted pairing connection. The one-time process writes the token to `/home/pi/.config/rob-vision/token` with owner-only permissions and attempts to restart the receiver. **Check Link** shows online when the receiver's authenticated polls reach the UNO Q; it turns offline after three seconds without a poll. Pairing supplies the token to an already prepared receiver; it does not install RetroArch mappings or hooks.
2. **Aim the camera.** Attach the camera to the UNO Q and select **Start Camera Check**. The live image shows the green sampling rectangle. Point it at the game's flashing area and check the delivered frame rate and signal level. If

the camera disconnects or stops delivering frames, reconnect the cable and select **Reset & Reconnect**; this closes the old capture, scans devices again, and restarts the camera in software. It does not reset the shared USB hub. Select **Stop Camera Check** before manual Gyromite gate tests. The attached Kiyo Pro delivered about 60 fps at 640×480 after its temporary HDR-off setting was applied. Both games' Test signals were recognized with a tight crop. Camera movement reception was intermittent; the installed RetroPie game-frame link now handles movement commands when active.

3. **Check Test mode.** Select Gyromite or Stack-Up and enter that game's Test mode. With the camera running, select **Watch Test Signal**. Gyromite's sustained green field or Stack-Up's alternating green/white Test signal makes the virtual R.O.B.'s red head light blink on Setup and Mission. A separately decoded ready-light command makes it glow steadily. Neither moves his arms or game pieces. **Preview Red Light** shows the blinking animation for a few seconds without claiming a camera signal was detected. If the page stays on Watching, adjust the camera and retry. The [Gyromite manual](#) and [Stack-Up manual](#) describe Test mode for adjusting R.O.B. to the screen. Both Test modes have been confirmed with the attached camera. In Stack-Up, press Select at the ROBOT BLOCK artwork to show the mode list, choose Test, then press Start. The Test signal appears over the ROBOT BLOCK artwork. Stack-Up CLOSE and DOWN commands have been verified through the RetroPie game-frame link.
4. **Try the controls.** For Gyromite, **Lower Red** and **Lower Blue** press the matching virtual pads immediately; select the same control again to raise it or **Release Both**. The automatic 60-second release still applies. For Stack-Up, **Left, Right, Up, Down, Open, Close** send one manual command to the virtual model. Stack-Up has no Gyromite-style Controller 2 pad return path. Check the Mission page's Live Model and Game Table for the resulting movement.

A local file:// Setup page is only a preview; use the UNO Q URL for live status. Setup shows delivered fps averaged over four seconds. Movement commands remain unverified on the actual display.

The Mission page keeps Fast Gates for responsive play. Its Game Table now sits beside Pose Preview and System Vitals. The camera image appears only on Setup.

CHAPTER 04 / CURRENT REFERENCE

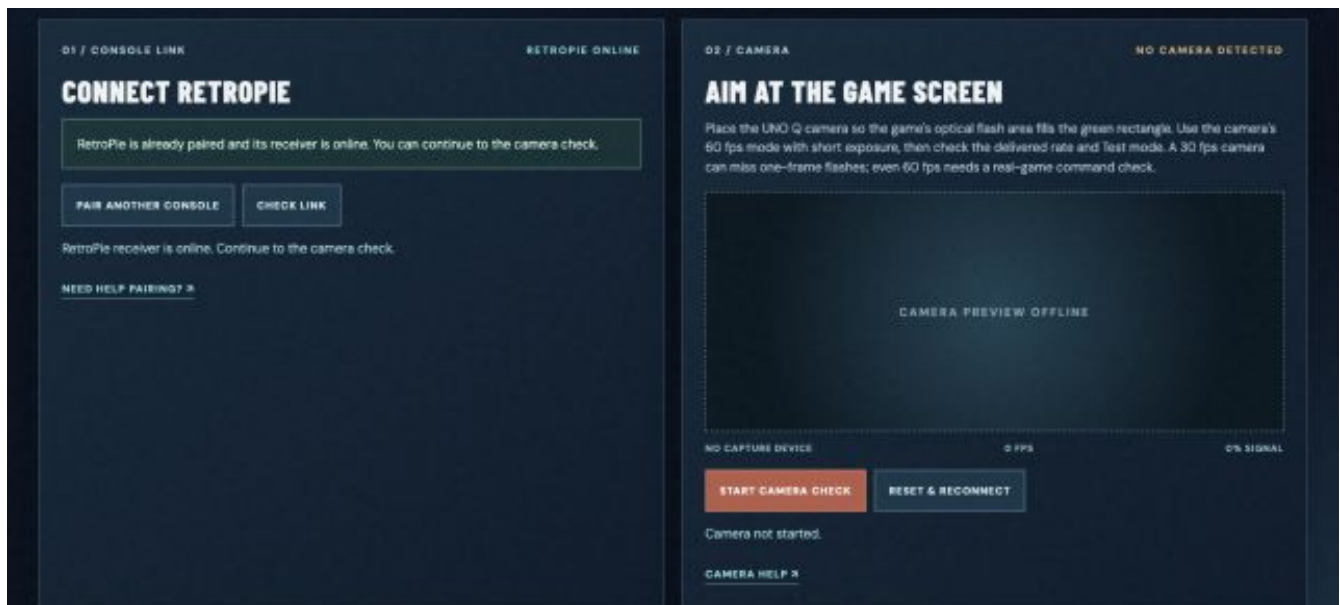
R.O.B. Vision installation guide

Installed test setup, 25 September 2026: R.O.B. Vision is a separate UNO Q App Lab app. `retropie.local` has exact-name runcommand hooks, pairing, and a virtual Controller 2 receiver. Gyromite launch/exit and both gate colors have been verified under FCEUmm. The UNO Q camera recognized both games' Test signals; camera-only movement remains intermittent. FCEUmm drove all six commands in both games through the frame link. Nestopia's frame link is installed and delivered live movement commands from both games to the UNO Q; its Gyromite Player 2 gate return remains to be checked. There is no physical robot to assemble.

UNO Q and browser

Start **R.O.B. Vision** under App Lab **My Apps**. This UNO Q runs one App Lab app at a time, so stop VirtualGlove first. App Lab exposes port 80 at `http://arduiain.local/dashboard/` and the controller directly at `http://arduiain.local:8766/dashboard/`. Both use one virtual game state. The App Lab Python image includes OpenCV; its `python/main.py` gateway talks to `controller/service.py`, and Router Bridge talks to the matrix sketch. Existing Avahi supplies the `.local` hostname. Do not enable the separate `deploy/rob-vision.service` while App Lab uses port 8766.

Open **Setup** to check pairing and frame a camera. Browser controls are available on the trusted LAN without a token prompt. The token in the UNO Q's `~/.config/rob-vision/environment` is for RetroPie launch and receiver identity. Keep the LAN private. A local `file://` copy is the independent preview; use the UNO Q URL for live status.



RetroPie

See [RetroPie deployment instructions](#) for hook files, root uinput receiver service, pairing, both FCEUmm and Nestopia frame-link choices, and RetroArch port 2 configuration. The current test machine already has these installed. On another machine, merge launch/end hooks with any existing custom scripts, then pair from the Setup page. **Check Link** becomes online after authenticated receiver polls. The receiver also scans the active RetroArch process so it can resync game context after a UNO Q restart.

For either supported game, choose `lr-robvision-fceumm` or `lr-robvision-nestopia` in RetroPie's emulator selection to keep automatic game-frame movement. These entries run the installed FCEUmm or Nestopia core through the R.O.B. Vision frame wrapper; they are not separate emulators. Choosing the plain `lr-fceumm` or `lr-nestopia` entry runs the game without that frame link. The UNO Q camera and manual controls are separate input paths.

Camera commissioning

Attach a compatible camera to the UNO Q, aim it at the game's optical flash region, and use Setup's **Start Camera Check**. Confirm that OpenCV found a real capture device, the green crop covers the flashing region, and measured fps is near 60. Test both games' Test mode, then record complete valid commands and false-trigger behavior on the intended screen. The current service asks for 60 fps and uses the center 60% crop by default. Even delivered 60 fps does not guarantee every one-frame flash; at 30 fps the synthetic decoder rejected them. The installed RetroPie FCEUmm and Nestopia frame-link choices read each rendered NES frame and handle automatic game commands without depending on the camera. Both delivered live commands from Gyromite and Stack-Up. Camera-only play remains experimental.

Reset & Reconnect retries capture in software. UNO Q camera-recovery, shutdown, early-start, and network-status host helpers are included in [deploy/uno-q](#), but are not installed by App Lab and need privileged host setup. The existing Ethernet adapter shares the USB hub, so the recovery helper guards against resetting a hub carrying network traffic.

Local development trial

From the repository, install `requirements-camera.txt` if camera capture is needed, run `python3 -m controller.service`, and open `http://127.0.0.1:8766/dashboard/`. For LAN access use `ROB_VISION_TOKEN` and `--host 0.0.0.0`; the token remains for RetroPie operations, while trusted-LAN browser controls are open. Use `--camera-index` and `--camera-roi x,y,width,height` for a chosen camera/crop. The static preview can be served with `python3 -m http.server 8000` and has no game link.

Run the [verification plan](#) before treating optical play as complete.

CHAPTER 05 / CURRENT REFERENCE

R.O.B. Vision troubleshooting

First check which page is open: `http://arduino.local/dashboard/` is live; a `file://` page or a simple static web server shows the independent preview.

Symptom	Check
App does not start, port 8766 in use	Stop VirtualGlove or the separate <code>rob-vision.service</code> ; App Lab and that service cannot both bind the port.
GYROMITE / CONNECTED but no robot movement	Check RETROPIE ONLINE and GAME FRAMES LINKED separately. If the latter is absent, restart the game so RetroPie loads the R.O.B. Vision core. Check Mission's Activity Feed for decoded or blocked actions.
RetroPie is online but no game is selected	A receiver can be online with no active game. Launch a registered ROM; then reload Mission if needed.
No camera or black preview	Check physical camera connection, device selection, exposure, and Setup's green sampling rectangle. Reset & Reconnect retries OpenCV; it does not reset USB power.
Test light never blinks	Enter the game's Test mode, select Watch Test Signal , align the crop, and check measured fps and brightness. A red-light preview is only an animation demonstration.
Movement command is missed	On RetroPie, confirm GAME FRAMES LINKED and the correct game is selected. The frame link reads every rendered NES frame; check the Activity Feed for a decoded command or a model move blocked by height, grip, or station. Camera-only timing remains a separate experiment.
Red/blue gate does not move	Confirm Gyromite is the active RetroArch game, Setup Check Link is online, camera is stopped for Fast Gates, and RetroArch port 2 is configured. Inspect the game screen during an isolated color hold.
Fast Gates vanished	They show only in live Gyromite with camera capture stopped.
Gate remains down	Select Release Both ; a hold also expires at 60 seconds. The receiver releases on game exit or stale network data.
Stack-Up block does not move	Review current station, height, gripper, carried group, and destination capacity; invalid moves leave the stack intact and appear in activity.
Demo does not start	Exit the running game first. Demo mode works when paired and idle; Home resets it.
Browser shows stale scene	Reload for a fresh <code>/api/state</code> snapshot. A local preview cannot show UNO Q state.

On the dedicated RetroPie test setup, `systemctl status rob-vision-controller2.service` checks the root virtual-controller service. Inspect RetroPie's `runcommand` log and configured ROM basename if launch identity is missing. `/dev/input/js1` is expected for the virtual Controller 2 while the player's controller is generally `/dev/input/js0`. Keep tokens and ROM files out of bug reports.